EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction; Basic definition ; Amorphous Solids; Isotropy; Anisotropy; Solid Properties

1. Which of the following statements is incorrect about amorphous solids?
(A) They are anisotropic
(B) They are rigid and incompressible
(C) They melt over a wide range of temperature
(D) There is no orderly arrangement of particles **CSS093**

2. Which of the following is/are pseudo solids?
I. KCl II. Barium chloride dihydrate
III. Rubber IV. Solid cake left after distillation of coal tar
(A) I, III (B) II, III (C) III, IV (D) III
CSS102

Classification of Crystalline Solids: molecular; metallic ; covalent and network solid

3. Which of the following solids are not correctly matched with the bonds found between the constituent particles :
(A) Solid CO_2 : Vander Waal's (B) Graphite : Covalent and Vander Waal
(C) Grey Cast Iron : Ionic (D) Metal alloys : ions-delocalised electrons
CSS099

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4. Which of the following is not correct for ionic crystals :
(A) They possess high melting point and boiling point
(B) All are electrolyte
(C) Exhibit the property of isomorphism
(D) Exhibit directional properties of the bond
CSS100

Lattice; Unit cell; Primitive Unit Cell; Seven Crystal Systems; Unit Cell shapes; Bravais lattices

5. A match box exhibits -
(A) Cubic geometry (B) Monoclinic geometry
(C) Tetragonal geometry (D) Orthorhombic geometry
CSS095

6. Crystal system in which maximum number of Bravais lattices are possible is
(A) Cubic (B) Triclinic (C) Orthorhombic (D) Rhombohedral
CSS096

7. The most unsymmetrical crystal system is :
(A) Cubic (B) Triclinic (C) Orthorhombic (D) Rhombohedral
CSS097

8. The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is:
(A) 7 (B) 5 (C) 2 (D) 3
CSS098

For More Join: @IITJEE_Advanced

9. The smallest repeating pattern which when repeated in three dimensions results in the crystal of the substance is called

(A) Space lattice (B) Crystal lattice

(C) Unit cell (D) coordination number

CSS101

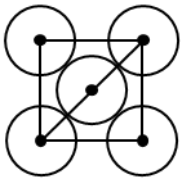
Packing in solids 1D; (One Dimension); 2D;(Two Dimension); Square Closed Packing; Hexagonal Closed Packing

10. Packing fraction in 2-D hexagonal arrangement -

(A) $\frac{\pi}{3\sqrt{2}}$ (B) $\frac{\pi}{3\sqrt{3}}$ (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi}{6}$

CSS110

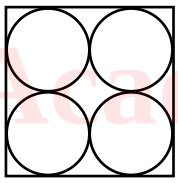
11. The packing efficiency of the two-dimensional square unit cell shown below is



(A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

CSS111

12. Identical 4 spheres are taken and are arranged in a layer of square packing touching each other as shown



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The percentage of vacant space is

(A) $100\left(1-\frac{3\pi}{8}\right)$ (B) $100\left(1-\frac{\pi}{6}\right)$ (C) $100-\frac{3\pi}{8}$ (D) $\frac{\pi}{6}$

CSS112

Packing in solids 3D; Three Dimension; Simple Cubic Closed Packing; SCC; Primitive Cubic; Density

13. The correct relation between edge length (a) and radius of atom (r) in simple cubic unit cell is –

(A) a = r (B) a = 2r (C) a = $\frac{r}{2}$ (D) None of these

CSS105

14. The packing efficiency of simple cubic unit cell is-

(A) 0.68 (B) 0.74 (C) 0.52 (D) 0.32

CSS106

15. The second coordination number of an atom in simple cubic unit cell is-

(A) 4 (B) 6 (C) 8 (D) 12

CSS107

16. What percentage of space is vacant in simple cubic unit cell ?

(A) 32% (B) 48% (C) 18% (D) None of these

CSS108

For More Join: @IITJEE_Advanced

17. The third coordination number of atom in simple cubic unit cell is-
(A) 4 (B) 6 (C) 8 (D) 12

CSS109

Body Centred Tetragonal(BCT); Body Centred Cubic Unit Cell(BCC)

18. Number of atoms at second nearest position from a given atom in a BCC structure is
(A) 8 (B) 6 (C) 12 (D) 4

CSS115

19. Lithium forms body centred cubic structure. The length of the side of its unit cell is 351 pm. Atomic radius of the lithium will be :-
(A) 152 pm (B) 75 pm (C) 300 pm (D) 240 pm

CSS116

20. A cubic solid is made up of two elements 'A' and 'B'. Atoms 'B' are at the corners of the cube and 'A' at the body centre. The simplest formula of compound
(A) AB (B) A_2B (C) AB_2 (D) A_2B_3

CSS117

21. In the body-centred cubic unit cell & face centred cubic unit cell, the radius of atom in terms of edge length (a) of the unit cell is respectively:
(A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$ (C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$

CSS118

Face Centred Cubic Unit Cell(FCC); Cubic Closed Packing(CCP)

22. Percentage area of each face covered by atoms in an FCC unit cell is -
(A) 60.4% (B) 68% (C) 74% (D) 78.5%

CSS119

23. Correct sequence of the coordination number in SCC, FCC & BCC is-
(A) 6, 8, 12 (B) 6, 12, 8 (C) 8, 12, 6 (D) 8, 6, 12

CSS120

24. In which of the following arrangement distance between two nearest neighbours is maximum, considering identical sized atoms in all arrangements ?
(A) Simple cubic (B) BCC (C) FCC (D) equal in all

CSS121

25. The ratio of densities if same element undergoes FCC as well as simple cubic packing will be (assume same atomic radius in both crystals)
(A) $\sqrt{2}:1$ (B) $1:\sqrt{2}$ (C) 1 : 1 (D) $\sqrt{2}:3$

CSS122

26. Platinum crystallizes in a face centred cubic crystal with a unit cell length 'a'. The distance between nearest neighbours is:
(A) a (B) $\frac{a\sqrt{3}}{2}$ (C) $\frac{a\sqrt{2}}{2}$ (D) $\frac{a\sqrt{2}}{4}$

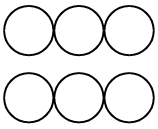
CSS123

27. All noble gases crystallise in the CCP structure except
(A) Helium (B) Neon (C) Argon (D) Krypton

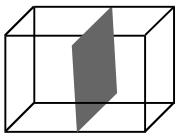
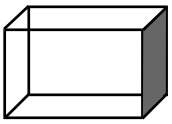
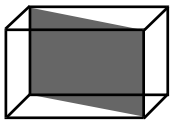
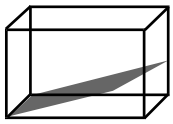
CSS133

For More Join: @IITJEE_Advanced

28. Which of the following shaded plane in FCC lattice contains arrangement of atoms as shown by circles :



Figure

(A)  (B)  (C)  (D) 

CSS135

29. A substance A_xB_y crystallises in an FCC lattice in which atoms "A" occupy each corner of the cube and atoms "B" occupy the centres of each face of the cube. Identify the correct composition of the substance A_xB_y.

(A) AB_3 (B) A_4B_3

(C) A_3B (D) composition cannot be specified

CSS167

Hexagonal Closed Packing(HCP)

30. The coordination number of a metal crystallizes in an HCP structure is

(A) 12 (B) 4 (C) 8 (D) 6

CSS129

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31. What is not true regarding hexagonal close packing (HCP)-

(A) packing fraction is 0.74

(B) coordination number is 12

(C) ABC ABC.....type packing

(D) Containing both tetrahedral and octahedral voids

CSS130

32. Which one of the following schemes of ordering closest packed sheets of equal sized spheres do not generate close packed lattice.

(A) ABCABC (B) ABACABAC (C) ABBAABBA (D) ABCBCABCBC

CSS131

33. HCP crystal is made of A ions, and B ions are arranged in $2/3^{rd}$ of octahedral voids. The simplest formula of compound is

(A) A_4B_6 (B) A_6B_4 (C) A_2B_3 (D) A_3B_2

CSS132

34. The ABAB close packing and ABC ABC close packing are respectively called :

(A) hexagonal close packing (hcp) and cubic close packing (ccp)

(B) ccp and hcp

(C) body centred cubic (bcc) packing and hexagonal close packing (hcp)

(D) hcp and bcc

CSS134

For More Join: @IITJEE_Advanced

Voids

35. The size of an octahedral void formed in a closest packed lattice as compared to tetrahedral void is
(A) Equal (B) Smaller (C) Larger (D) Not definite

CSS141

36. Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm. The radius of largest atom that can fit into the voids of copper lattice without disturbing it is
(A) 53 pm (B) 45 pm (C) 93 pm (D) 87 pm

CSS142

37. In FCC unit cell smallest distance between octahedral void & tetrahedral void is -
(a = edge length of unit cell)
(A) $\frac{a}{\sqrt{2}}$ (B) $\frac{\sqrt{3}a}{2}$ (C) a (D) $\frac{\sqrt{3}a}{4}$

CSS143

38. A cubic solid is made by atoms 'A' forming close pack arrangement, 'B' occupying one-fourth of tetrahedral void and C occupying half of the octahedral voids. What is the formula of compound ?
(A) A_2BC (B) AB_2C (C) ABC_2 (D) ABC

CSS144

39. For ABC ABC ABC .... packing, distance between two successive tetrahedral voids is X and distance between two successive octahedral voids is y in a unit cell, then $\frac{y\sqrt{2}}{x}$ is
(A) 1 (B) 2 (C) 3 (D) 4

CSS145

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40. An element 'M' crystallizes in ABAB....type packing. If adjacent layer A & B are $10\frac{\sqrt{2}}{\sqrt{3}}$ pm apart, then radius of largest sphere which can be fitted in the void without disturbing the lattice arrangement will be (Given : $\sqrt{2}$ =1.414)
(A) 1.7 pm (B) 2.9 pm (C) 2.07 pm (D) 3.5 pm

CSS146

41. Distance between two nearest tetrahedral voids in FCC unit cell is -
(A) $\sqrt{2}a$ (B) $\frac{a}{\sqrt{2}}$ (C) $\frac{a}{2}$ (D) $\frac{a}{\sqrt{3}}$

CSS147

42. If a unit cell of FCC lattice is divided into eight identical small cubes, then
(A) Each small cube contains one octahedral void.
(B) Every alternate small cube contains one octahedral void.
(C) Each small cube contains one tetrahedral void.
(D) Every alternate small cube contains one tetrahedral void.

CSS151

Radius Ratio

43. If the radius of $Mg^{2\oplus}$ ion, $Cs^{\oplus}$ ion, $O^{2\ominus}$ ion, $S^{2\ominus}$ ion and $Cl^{\ominus}$ ion are 0.65 Å, 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively, the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl are respectively
(A) 4, 6, 8 (B) 6, 4, 8 (C) 8, 6, 4 (D) 8, 4, 6

CSS148

For More Join: @IITJEE_Advanced

44. The two ions $A^{\oplus}$ and $B^{\ominus}$ have radii 88 pm and 200 pm respectively. In the closed packed crystal of compound AB, the co-ordination number of $A^{\oplus}$
(A) 2 (B) 4 (C) 6 (D) 8

CSS149

45. An ionic compound AB has ZnS type structure. If the radius of $A^{\oplus}$ is 22.5 pm, then the ideal radius of $B^{\ominus}$ would be
(A) 54.35 pm (B) 100 pm (C) 145.16 pm (D) none of these

CSS150

Packing in Ionic Solids: NaCl; ZnS; CsCl; Na_2O ; CaF_2 structure

46. In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is
(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4

CSS166

47. Which of the following FCC structure contains cations in alternate tetrahedral voids?
(A) NaCl (B) ZnS (C) Na_2O (D) CaF_2

CSS168

48. Which of the following metal oxide is anti-ferromagnetic in nature?
(A) MnO (B) TiO_2 (C) VO_2 (D) CrO_2

CSS175

Imperfection in Solid: Defects in Solids

49. Which of the following is the most likely to show Schottky defect ?
(A) CaF_2 (B) ZnS (C) AgCl (D) CsCl

CSS153

50. In the Schottky defect, in AB type ionic solids
(A) cations are missing from the lattice sites and occupy the interstitial sites
(B) equal number of cations and anions are missing
(C) anions are missing and electrons are present in their place
(D) equal number of extra cations and electrons are present in the interstitial sites

CSS154

51. The measured density of AgI is 6.94 g/cm^3 and the theoretical density is 5.67 g/cm^3. These data indicate that solid AgI has -
(A) Schottky defect (B) Frenkel defect
(C) Interstitial impurities defect (D) Both (A) and (B)

CSS155

52. Which of the following statement is CORRECT ?
(A) A metal can show only non- stoichiometric defects
(B) Schottky defect reduces the density of a solid due to significant increase in volume.
(C) Impurity defect always change the density.
(D) Solids having F-centres may have metal excess defect due to missing anions.

CSS156

53. Statement-1 : Due to Frenkel defect, there is no effect on the density of the crystalline solid.
Statement-2 : In Frenkel defect, no cation or anion leaves the crystal.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CSS157

For More Join: @IITJEE_Advanced

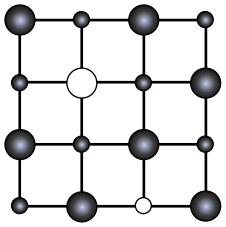
54. Due to Frenkel defect, the density of the ionic solids
(A) Increases (B) Decreases (C) Does not change (D) Changes

CSS158

55. Which of the following has Frenkel defect?
(A) Sodium chloride (B) Graphite (C) Silver bromide (D) Diamond

CSS159

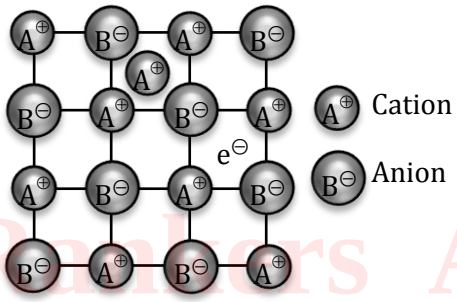
56. The structure shown here represents



(A) Schottky defect (B) Frenkel defect (C) Metal excess defect (D) None

CSS160

57. The structure shown here represents



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(A) Schottky defect
(B) Frenkel defect
(C) Metal excess defect because of absent anion
(D) Metal excess defect because of excess cation

CSS161

58. In AgCl, the $Ag^{\oplus}$ ions are displaced from their lattice position to an interstitial position. Such a defect is called
(A) Schottky defect (B) Frenkel defect (C) F-centre defect (D) Metal excess defect

CSS162

59. NaCl shows Schottky defect and AgCl shows Frenkel defect. Their electrical conductivity is due to the
(A) Motion of electrons and not the motion of ions
(B) Motion of ions and not the motion of electrons
(C) Lower coordination number of NaCl
(D) Higher coordination number of AgCl

CSS163

60. Due to Frenkel defect
(A) Density of the crystal decreases (B) Conductivity increases
(C) Conductivity decreases (D) Crystal becomes charged electrically

CSS164

For More Join: @IITJEE_Advanced

61. A metal crystallizes into two cubic phases, FCC and BCC whose unit cell lengths are 3.5 and 3.0 Å respectively. Calculate the ratio of densities of FCC and BCC.

(A) 1.259 (B) 1.558 (C) 1.359 (D) 1.650

CSS165

Electrical properties

62. If we mix a pentavalent impurity in a crystal lattice of germanium, what type of semiconductor formation will occur?

(A) p-type (B) n-type (C) Both (A) and (B) (D) None of these

CSS169

63. Silicon doped with group 13 and group 15-member elements is, respectively, called ..... semiconductor

(A) p-type, n-type (B) n-type, p-type (C) p-type (D) n-type

CSS170

64. The electrical conductivity of semiconductor is

(A) 10^{8} ohm^{-1} cm^{-1} (B) 10^{-22} ohm^{-1} cm^{-1}

(C) In the range of 10^{-9} to 10^{2} ohm^{-1} cm^{-1} (D) None of the above

CSS171

65. Silicon and germanium are

(A) Conductors (B) Insulators (C) Semiconductors (D) None of the above

CSS172

66. To get n-type doped semiconductor, impurity to be added to silicon should have the following number of valence electrons

(A) 2 (B) 5 (C) 3 (D) 1

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CSS173

67. A semiconductor of Ge can be made p-type adding

(A) Trivalent impurity (B) Tetravalent impurity

(C) Pentavalent impurity (D) Divalent impurity

CSS174

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Introduction; Basic definition ; Amorphous Solids; Isotropy; Anisotropy; Solid Properties

1. Which of the following statements is/are false.

(A) It is the isotropic nature of amorphous solids that all the bonds are of equal strength.

(B) Entropy of a crystalline solid is higher than that of an amorphous solid.

(C) Amorphous solids have definite volume but not definite shape.

(D) Amorphous solids can become crystalline on slow heating.

CSS104

2. Which of the following features are shown by quartz glass

(A) This is a crystalline solid

(B) This has definite Heat of fusion

(C) This is also called super cooled liquid

(D) Refractive index is same in all the directions.

CSS094

3. Which of the following solid substance(s) will have the same refractive index when measured in different directions?

(A) Rubber (B) NaCl (C) Plastic (D) Graphite

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CSS280

Body Centred Tetragonal(BCT); Body Centred Cubic Unit Cell(BCC)

4. In a body centred unit cell containing only one type of atom. (R = Radius of atom)

(A) The edge length is equal to $\frac{4R}{\sqrt{3}}$.

(B) The edge length is equal to $\sqrt{2R^2+2R^2}$.

(C) The edge length is equal to $\frac{2d}{\sqrt{3}}$, where d= nearest neighbour distance.

(D) Shortest distance between next nearest neighbour atoms is $\frac{4R}{\sqrt{3}}$.

CSS124

Hexagonal Closed Packing(HCP)

5. The HCP and CCP structure of a given element. (Given radius of element is same in both structures)

(A) have same density

(B) have same distance between two consecutive layers (A & B)

(C) have same co-ordination number

(D) have same fraction of unoccupied space.

CSS137

For More Join: @IITJEE_Advanced

6. Which of the following is/are true ?

(A) Ratio of nearest neighbours in simple cubic cell to next nearest neighbours in face centred cubic cell is 1.

(B) Packing efficiency of a unit cell in which atoms are present at each corner and each edge centre is about 26 % in metallic crystal.

(C) Distance between two planes in FCC or HCP arrangement is same for a metal existing in both forms, with same atomic radius.

(D) If number of unit cell along one edge are 'x', then total number of unit cell in cube = x^3 .

CSS176

Voids

7. In which of the following arrangement regular octahedral voids are formed?

(A) BCC (B) FCC (C) HCP (D) SC

CSS152

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For More Join: @IITJEE_Advanced

EXERCISE (S)

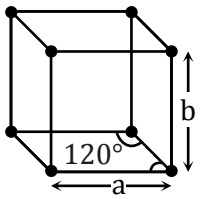
Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Lattice; Unit cell; Primitive Unit Cell; Seven Crystal Systems; Unit Cell shapes; Bravais lattices

1. Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants where **a** = 5 Å, and **b** = $3.5\sqrt{3}$ Å. How many molecules are contained in the given unit cell?

[density (ice) = $\frac{16}{17.5}$ gm/cm^3)]



CSS183

Packing in solids 1D; One Dimension; 2D; Two Dimension; Square Closed Packing; Hexagonal Closed Packing

2. Marbles of diameter 10 mm each are to be arranged on a flat surface so that their centres lie within the area enclosed by four lines of length each 40 mm. Sketch the arrangement that will give the maximum number of marbles per unit area, that can be enclosed in this manner and deduce the expression to calculate it.

CSS113

Body Centred Tetragonal(BCT); Body Centred Cubic Unit Cell(BCC)

3. Potassium has body-centred cubic structure with the nearest neighbour distance $260\sqrt{3}$pm. Its density (in kg/m^3) would be

($\frac{1}{(5.2)^3}$ = 0.007, $N_A = 6 \times 10^{23}$, K = 39)

CSS128

Face Centred Cubic Unit Cell(FCC); Cubic Closed Packing(CCP)

4. The density of diamond from the fact that it has face centred cubic structure with two atoms per lattice point and unit cell edge length of 3.6 Å, is 'x' gm / cm^3, then the value of (1.458x) is ($N_A = 6 \times 10^{23}$)

CSS178

5. Iron crystallizes in several modifications. At about 910°C, the body-centred cubic 'α' form undergoes a transition to the face-centred cubic 'γ' form. Assuming that the distance between nearest neighbours is the same in the two forms at the transition temperature, the ratio of the density of γ iron to that of α iron at the transition temperature, x : 1, then the value of $(3\sqrt{6}x)$ is.

CSS179

For More Join: @IITJEE_Advanced

Hexagonal Closed Packing (HCP)

6. Element A crystallizes in HCP. Total number of tetrahedral voids in 24 micrograms of A are 6×10^{x}, the value of x ($N_A = 6 \times 10^{23}$, Atomic mass of A = 48)

CSS139

Radius Ratio

7. An element (atomic weight = 125) crystallises in simple cubic structure. Diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. If the density of element 'x' gm/cm^3, the value of $\left(\frac{6x}{5}\right)$ is -

[**Given** : = 1.732 , $N_A = 6 \times 10^{23}$]

CSS177

Packing in Ionic Solids: NaCl; ZnS; CsCl; Na_2O ; CaF_2 structure

8. Calculate the density (in gm/cm^3) of NaCl type ionic solid (Molar mass = 75 gm/mol) if distance between two nearest cations is $250\sqrt{2}$ pm (Avogadro number = 6×10^{23})

CSS189

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

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Packing in solids 3D; Three Dimension; Simple Cubic Closed Packing; SCC; Primitive Cubic; Density

9. An element (Given mass–288 g) crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two extra atoms on one of its body diagonals. If the volume of this unit cell is 2.4×10^{-23} cm^3 and density of element is 7.2 g cm^{-3}, then the number of atom is x $\times 10^{24}$, then the value of x will be.

CSS281

Face Centred Cubic Unit Cell(FCC); Cubic Closed Packing(CCP)

10. The effective radius of the iron atom is $\sqrt{2}$ Å. It has FCC structure. Calculate its density. (Fe = 56 amu, $N_A = 6 \times 10^{23}$)

CSS126

11. In FCC unit cell, what fraction of edge is not covered by atoms?

CSS127

12. An element X (atomic weight = 24 gm/mole) forms a face centred cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is 2.40×10^3 kg/m^3, calculate the percentage occupancy of lattice points by X element. (Given : $N_A = 6 \times 10^{23}$)

CSS188

Hexagonal Closed Packing(HCP)

13. Find packing fraction of three-dimensional unit cell of AAAAA.....type hypothetical arrangement in which hexagonal packing is taken in layer.

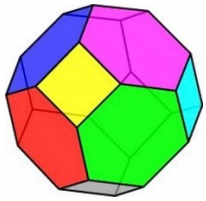
CSS140

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The number of hexagonal faces that present in a truncated octahedron is.

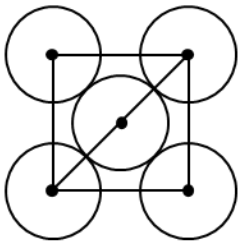
[JEE (Advanced) 2011]



CSS084

2. The packing efficiency of the two-dimensional square unit cell shown below is

[JEE (Advanced) 2010]



(A) 39.27% (B) 68.02% (C) 74.05% (D) 78.54%

CSS083

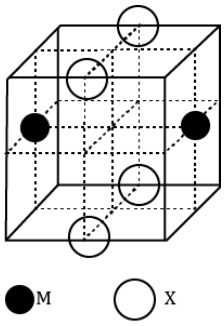
3. The edge length of unit cell of a metal having atomic weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm.

[JEE (Advanced) 2006]

CSS077

4. The cubic unit cell structure of a compound containing cation M and anion X is shown below. When compared to the anion, the cation has smaller ionic radius. Choose the correct statement(s).

[JEE (Advanced) 2020]



(A) The empirical formula of the compound is MX.
(B) The cation M and anion X have different coordination geometries.
(C) The ratio of M-X bond length to the cubic unit cell edge length is 0.866.
(D) The ratio of the ionic radii of cation M to anion X is 0.414.

CSS090

For More Join: @IITJEE_Advanced

5. Atoms of metals x, y, and z form face-centred cubic (fcc) unit cell of edge length L_x, body-centred cubic (bcc) unit cell of edge length L_y, and simple cubic unit cell of edge length L_z, respectively.

If $r_z = \frac{\sqrt{3}}{2} r_y ; r_y = \frac{8}{\sqrt{3}} r_x ; M_z = \frac{3}{2} M_y$ and $M_z = 3M_x$, then the correct statement (s) is (are)

[Given : M_x, M_y, and M_z are molar masses of metals x, y, and z, respectively.
r_x, r_y, and r_z are atomic radii of metals x, y, and z, respectively.]

[JEE (Advanced) 2023]

(A) Packing efficiency of unit cell of x > Packing efficiency of unit cell of y > Packing efficiency of unit cell of z

(B) $L_y > L_z$

(C) $L_x > L_y$

(D) Density of x > Density of y

CSS282

6. Atom X occupies the fcc lattice sites as well as alternate tetrahedral voids of the same lattice. The packing efficiency (in %) of the resultant solid is closest to

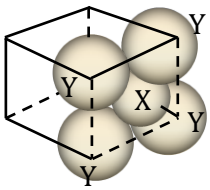
[JEE (Advanced) 2022]

(A) 25 (B) 35 (C) 55 (D) 75

CSS092

7. For the given close packed structure of a salt made of cation X and anion Y shown below (ions of only one face are shown for clarity), the packing fraction is approximately

(packing fraction = $\frac{\text{Packing efficiency}}{100}$) **[JEE (Advanced) 2021]**



(A) 0.74 (B) 0.63 (C) 0.52 (D) 0.48

CSS091

8. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8g cm^{-3}, then the number of atoms present in 256g of the crystal is N × 10^{24}. The value of N is **[JEE (Advanced) 2017]**

CSS088

9. The **CORRECT** statement(s) for cubic close packed (CCP) three-dimensional structure is (are)

[JEE (Advanced) 2016]

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12

(B) The efficiency of atom packing is 74%

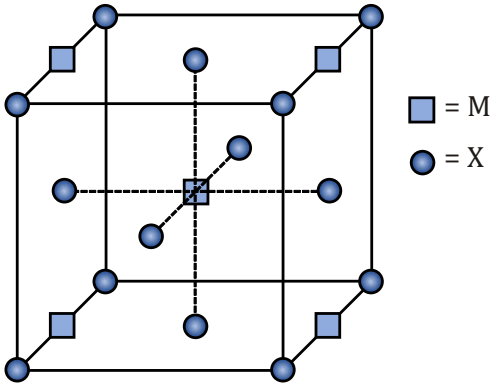
(C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

CSS087

For More Join: @IITJEE_Advanced

10. A compound M_pX_q has cubic close packing (CCP) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is : **[JEE (Advanced) 2012]**



(A) MX (B) MX_2 (C) M_2X_4 (D) M_5X_{14}

CSS085

11. An element crystallises in FCC lattice having edge length 400 pm. Calculate the maximum diameter which can be placed in interstitial sites without disturbing the structure.

[JEE (Advanced) 2005]

CSS075

Hexagonal Closed Packing(HCP)

Paragraph for Question No. 12 to 14

Pankers Academy JEE

In hexagonal systems of crystals, a frequently encountered arrangement of atoms is described as a hexagonal prism. Here, the top and bottom of the cell are regular hexagons and three atoms are sandwiched in between them. A space-filling model of this structure, called hexagonal close-packed (HCP), is constituted of a sphere on a flat surface surrounded in the same plane by six identical spheres as closely as possible. Three spheres are then placed over the first layer so that they touch each other and represent the second layer. Each one of these three spheres touch three spheres of the bottom layer. Finally, the second layer is covered with a third layer that is identical to the bottom layer in relative position. Assume radius of every sphere to be 'r'.

12. The number of atoms in this HCP unit cells is **[JEE (Advanced) 2008]**

(A) 4 (B) 6 (C) 12 (D) 17

CSS079

13. The volume of this HCP unit cell is **[JEE (Advanced) 2008]**

(A) $24\sqrt{2}r^3$ (B) $16\sqrt{2}r^3$ (C) $12\sqrt{2}r^3$ (D) $\frac{64}{3\sqrt{3}}r^3$

CSS080

14. The empty space in this HCP unit cell is **[JEE (Advanced) 2008]**

(A) 74% (B) 47.6 % (C) 32% (D) 26%

CSS081

15. The coordination number of a metal crystallising in a HCP structure is **[JEE (Advanced) 2000]**

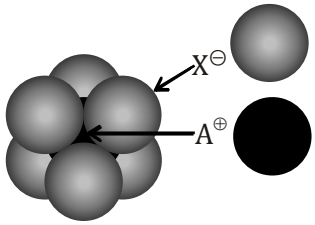
(A) 12 (B) 4 (C) 8 (D) 6

CSS072

For More Join: @IITJEE_Advanced

Radius Ratio

16. The arrangement of $X^{\ominus}$ ions around $A^{\oplus}$ ion in solid AX is given in the figure (not drawn to scale). If the radius of $X^{\ominus}$ is 250 pm, the radius of $A^{\oplus}$ is - **[JEE (Advanced) 2013]**



(A) 104 pm (B) 125 pm (C) 183 pm (D) 57 pm

CSS086

Packing in Ionic Solids: NaCl; ZnS; CsCl; Na_2O ; CaF_2 structure

17. Consider an ionic solid MX with NaCl structure. Construct a new structure (Z) whose unit cell is constructed from the unit cell of MX following the sequential instructions given below. Neglect the charge balance. **[JEE (Advanced) 2018]**

(i) Remove all the anions (X) except the central one

(ii) Replace all the face centred cations (M) by anions (X)

(iii) Remove all the corner cations (M)

(iv) Replace the central anion (X) with cation (M)

The value of $\left(\frac{\text{number of anions}}{\text{number of cations}}\right)$ in Z is___.

CSS089

Rankers Academy JEE

18. AB crystallizes in a rock salt structure with A : B = 1 : 1. The shortest distance between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu where Y is any arbitrary constant. Find the density in kg m^{-3}. **[JEE (Advanced) 2004]**

CSS074

19. In a solid "AB" having NaCl structure "A" atoms occupy the corners of the cubic unit cell. If all the face-centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is **[JEE (Advanced) 2000]**

(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4

CSS071

Imperfection in Solid: Defects in Solids

20. The correct statement(s) regarding defects in solid is (are) **[JEE (Advanced) 2009]**

(A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion.

(B) Frenkel defect is a dislocation defect

(C) Trapping of an electron in the lattice leads to the formation of F-center.

(D) Schottky defects have no effect on the physical properties of solids.

CSS082

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	C	D	D	C	B	D	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	B	B	C	D	B	C	B	A	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	D	B	D	A	C	A	C	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	D	A	C	A	D	A	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	C	A	C	B	D	B	A	D	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	D	A	C	C	A	D	B	B	B
Que.	61	62	63	64	65	66	67			
Ans.	A	B	A	C	C	B	A			

Rankers Academy JEE

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7
Ans.	ABC	CD	AC	ACD	ABCD	ABCD	BC

EXERCISE (S)

1.	4	**2.**	18	**3.**	910	**4.**	5	**5.**	8
6.	17	**7.**	2	**8.**	4	**9.**	5.00	**10.**	5.83
11.	0.293	**12.**	96.00	**13.**	0.604				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	D	216	AC	ABD	B	B	2	BCD	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	117.1	B	A	D	A	A	3	5	D	BC

For More Join: @IITJEE_Advanced